

**INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR**

1. Name of the Academic Unit: Bioscience and Biotechnology

2. Subject Name: IMMUNOLOGY LTP- 3-1-0,CRD- 4

3. Pre-requisites:

4. 1)BIOCHEMISTRY.

2)CELL AND MOLECULAR BIOLOGY.

3)MICROBIOLOGY

5. Syllabus and reference books: See the content

6. Text Book:

1. Janeway's Immunobiology 9th Edition by Kenneth M Murphy and Casey Weaver
ISBN : 978-0-815-34505-3

2. Kuby Immunology, 7th Edition by Judith A. Owen; Jenni Punt; Sharon A. Stranford
ISBN 13: 9781464119

7. Lecture-wise break-up:

Topic	Hour
The basic concept of immunology	4
primary and secondary lymphoid organ	1
Antibody Structure	2
molecular basis of antibody diversity	3
synthesis of antibody and secretion	1
Immunotechniques	3
major histocompatibility complex (MHC)	2
antigen processing and presentation	1
humoral immunity	2
B cell, development and immunity	3
complement	3
T cell and macrophages	2
cell mediated immunity	2
Cytokines: functions and mechanisms	2
Leukocyte migration and inflammation	1
immune tolerance	1
hyper sensitivity	1
autoimmunity	1
graft versus host reaction	1
Tutorial	12
CLASS Test	2
TOTAL	50